

Exercise 1: Foundations and Visual Encodings

Three worksheets, three captions. No memo, no screenshots.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv from MVU_Walkthrough_Data.zip (1,400 rows, one row per student)
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 1, Ch 3, and Ch 7 (Making Visualizations That Look Great and Work Well)
Walkthrough	Module 1 of the MVU Tableau Walkthrough follows these same steps, with checkpoints you can check as you go.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a visible caption on each. No memo, no screenshots.

Set up, connect, and fix the metadata

1. Connect to mvu_student_dim.csv (Connect > Text file).
2. On the data source page, confirm entry_date is a Date and undergrad_gpa is a Number (decimal).
3. On Sheet 1, drag student_id above the line into Dimensions. The sum of a set of ID numbers is not a quantity.

Checkpoint: How many rows does Tableau report at the bottom of the data source page?

Warm-up (not submitted)

Put college on Rows and student_id on Columns as Count (Distinct). Then drag cohort_year to Columns, see it arrive green (continuous), and switch it to Discrete (blue).

Checkpoint: How many bars appear with college on Rows?

Worksheet 1: bars and color

1. New worksheet. campus_name to Rows; student_id to Columns as Count (Distinct). Sort descending.
2. Drag enrollment_intensity to Color. This makes a stacked bar.
3. Duplicate the sheet. On the copy, also drag enrollment_intensity to Rows, to the right of campus_name, so the bars sit side by side.
4. Name the sheets Worksheet 1 — Stacked and Worksheet 1 — Side by Side.

Checkpoint: How many part-time students are at Mesquite Main Campus?

Worksheet 2: the scatter plot, and what it hides

1. New worksheet. credits_attempted to Columns, credits_earned to Rows. You get one mark: the grand total.
2. Drag student_id to Detail (one mark per student) and status to Color.
3. Analytics pane > drag Trend Line > Linear onto the view. Hover it for R-squared.
4. The chart holds 1,400 students. Count the dots you can actually see.

5. Fix the overplotting and keep the fix: lower opacity to about 30% and add mark borders, or take `student_id` off Detail and put `CNTD(student_id)` on Size.
6. Quick comparison: swap the axes for `undergrad_gpa` and `total_aid_awarded` and read R-squared again. You don't need to keep this version.

Checkpoint: R-squared of the linear trend line, credits attempted vs. credits earned (three decimals)?

Worksheet 3: the ethics of the axis

1. New worksheet. `campus_name` to Columns; `is_withdrawn` to Rows as Average. Set its default number format to Percentage, 1 decimal. Add labels.
2. Duplicate the sheet. On the copy, Edit Axis > untick Include zero and start the range at 12%. Name it Worksheet 3 — Truncated.
3. On the original, drag `enrollment_intensity` to Columns after `campus_name`, and to Color.
4. Apply Ch 7: remove gridlines and borders you don't need, and don't color by campus when the axis already names it.
5. Retitle the sheet with the finding, not the field names. Name it Worksheet 3 — Honest.

Checkpoint: Overall withdrawal rate at MVU Online, before the split (percentage, one decimal)?

Your three captions

Each worksheet carries a caption: one or two sentences answering one question, written while the view is in front of you. Turn it on with Worksheet > Show Caption, then double-click below the view to edit. Where a worksheet has two versions, put the caption on the second one.

Sheet	What your caption must answer
Worksheet 1	Which version makes it easier to compare part-time headcount across campuses, and why? Use the idea of a common baseline.
Worksheet 2	How many distinct dots can the reader actually see, and what did your fix reveal? What does R-squared tell you here, and what does it not establish?
Worksheet 3	Why is a zero baseline non-negotiable for a bar chart? After the split, write the finding a dean should read: one that doesn't blame the campus for something enrollment intensity explains.

Worksheet 3's caption carries 25% on its own. A caption saying "Online has the worst withdrawal rate" does not earn full marks. The split is the point of the exercise; the other two assess whether you can operate the software.

Before you submit: open each sheet and confirm the caption is showing. A caption toggled off still holds your text but won't be visible when your instructor opens the workbook.

Submit for Exercise 1

- Worksheet 1: campus headcount by enrollment intensity, stacked and side by side
- Worksheet 2: credits scatter with `student_id` on Detail, a trend line, and a visible overplotting fix
- Worksheet 3: withdrawal rate by campus, truncated and honest, with the honest version split by enrollment intensity
- A visible caption on each
- Packaged workbook (.twbx). No memo, no screenshots.